

To find following the machine learning regression method using in  $r^2$  value

1.MULTIPLE LINEAR REGRESSION ( $R^2$  value)=0.9358

2.SUPPORT VECTOR MACHINE

| S.No | HYPER<br>PARAMETER | LINEAR(r<br>value) | RBF (NON<br>LINEAR)(r<br>value) | POLY<br>(r value) | SIGMOID<br>(r value) |
|------|--------------------|--------------------|---------------------------------|-------------------|----------------------|
| 1    | C10                | -2.4365            | -0.0558                         | 0.0035            | 0.0575               |
| 2    | C100               | -357.0026          | -0.0310                         | 0.3858            | -0.0584              |
| 3    | C500               | -                  | 0.0469                          | 0.6155            | -0.0621              |
| 4    | C1000              | -                  | 0.1556                          | 0.6318            | -0.0668              |
| 5    | C2000              | -                  | 0.2856                          | 0.6472            | -0.0767              |
| 6    | C3000              | -                  | 0.3890                          | 0.6779            | -0.0870              |

The SVM Regression use  $R^2$  value ( (poly) and hyper parameter(C3000)) =0.6779

3.DECISION TREE

| SL.NO | CRITERION     | MAX FEATURES | SPLITTER | R VALUE |
|-------|---------------|--------------|----------|---------|
| 1     | squared_error | None         | best     | 0.9073  |
| 2     | squared_error | None         | randam   | 0.7434  |
| 3     | squared_error | sqrt         | best     | 0.8058  |
| 4     | squared_error | sqrt         | random   | 0.0278  |
| 5     | squared_error | Log2         | best     | 0.3082  |
| 6     | squared_error | Log2         | random   | 0.5897  |
| 7     | friedman_mse  | None         | best     | 0.9065  |
| 8     | friedman_mse  | None         | randam   | 0.9588  |
| 9     | friedman_mse  | sqrt         | best     | 0.1626  |
| 10    | friedman_mse  | sqrt         | random   | 0.5354  |

|    |                |      |        |         |
|----|----------------|------|--------|---------|
| 11 | friedman_mse   | Log2 | best   | 0.1623  |
| 12 | friedman_mse   | Log2 | random | 0.2129  |
| 13 | absolute_error | None | best   | 0.9627  |
| 14 | absolute_error | None | randam | 0.6981  |
| 15 | absolute_error | sqrt | best   | 0.8012  |
| 16 | absolute_error | sqrt | random | 0.4149  |
| 17 | absolute_error | Log2 | best   | 0.5436  |
| 18 | absolute_error | Log2 | random | 0.8194  |
| 19 | poisson        | None | best   | 0.9277  |
| 20 | poisson        | None | randam | 0.9104  |
| 21 | poisson        | sqrt | best   | 0.5605  |
| 22 | poisson        | sqrt | random | 0.5641  |
| 23 | poisson        | Log2 | best   | -0.2943 |
| 24 | poisson        | Log2 | random | -0.5544 |